

Part 2 — Test Case Documentation

Application Under Test: Factorial Calculator — <http://qainterview.pythonanywhere.com>

Testing Type: Functional, Boundary, Validation, UI, Navigation

Browser(s): Chrome, Microsoft Edge

OS: Windows 11

Total Test Cases: 26 **Passed:** 15 **Failed:** 11

Test Cases

ID	Test Case Description	Input	Test Steps	Expected Result	Actual Result	Status
TC01	Valid integer input	5	1. Enter 5 in the input field 2. Click Calculate	The factorial of 5 is: 120	The factorial of 5 is: 120	Pass
TC02	Negative number input	-5	1. Enter -5 in the input field 2. Click Calculate	Display error: "Factorial is not defined for negative numbers" or "Please enter a positive integer"	No response or feedback is displayed	Fail
TC03	Decimal input	2.5	1. Enter 2.5 in the input field 2. Click Calculate	Display error: "Please enter an integer"	Please enter an integer	Pass
TC04	Zero input	0	1. Enter 0 in the input field 2. Click Calculate	The factorial of 0 is: 1	The factorial of 0 is: 1	Pass
TC05	Blank input (no value)	(empty)	1. Leave input field blank 2. Click Calculate	Display error: "This field is required" or "Please enter a value"	Displays generic error: "Please enter an integer"	Fail
TC06	Large integer input	991	1. Enter 991 in the input field 2. Click Calculate	Display error: "Number too large to calculate"	The factorial of 991 is: Infinity	Fail
TC07	Very large integer input	1000	1. Enter 1000 in the input field 2. Click Calculate	Display validation message: "Number too large to calculate"	No output or error message displayed	Fail
TC08	Button processing feedback	5	1. Enter 5 2. Click Calculate and observe button state	Loading indicator or button disabled during processing	No processing feedback shown	Fail
TC09	Input field placeholder guidance	(any)	1. Click into the input field 2. Begin typing 3. Observe UI guidance	Persistent label or helper text remains visible	Placeholder disappears; no additional guidance shown	Fail
TC10	Boundary — minimum valid input (1)	1	1. Enter 1 in the input field 2. Click Calculate	The factorial of 1 is: 1	The factorial of 1 is: 1	Pass

ID	Test Case Description	Input	Test Steps	Expected Result	Actual Result	Status
TC11	Boundary — upper valid input (170)	170	1. Enter 170 in the input field 2. Click Calculate	The factorial of 170 is: 7.257415615307994e+306 (a valid large number)	Returns a valid large exponential result	Pass
TC12	Boundary overflow — first Infinity value (171)	171	1. Enter 171 in the input field 2. Click Calculate	Display error: "Number too large to calculate"	The factorial of 171 is: Infinity	Fail
TC13	Boundary overflow — upper range (991)	991	1. Enter 991 in the input field 2. Click Calculate	Display error: "Number too large to calculate"	The factorial of 991 is: Infinity	Fail
TC14	No response for input >= 992	992	1. Enter 992 in the input field 2. Click Calculate	Display error: "Number too large to calculate"	No output or error message displayed	Fail
TC15	Alphabetic input	abc	1. Enter abc in the input field 2. Click Calculate	Display error: "Please enter an integer"	Please enter an integer	Pass
TC16	Special characters input	!@#\$\$%	1. Enter !@#\$\$% in the input field 2. Click Calculate	Display error: "Please enter an integer"	Please enter an integer	Pass
TC17	Whitespace-only input	(spaces)	1. Enter only spaces in the input field 2. Click Calculate	Display error: "This field is required" or "Please enter an integer"	No response or generic error	Fail
TC18	Floating point that resembles integer (1.0)	1.0	1. Enter 1.0 in the input field 2. Click Calculate	Display error: "Please enter an integer" (1.0 is not an integer)	Please enter an integer	Pass
TC19	Negative decimal input	-1.5	1. Enter -1.5 in the input field 2. Click Calculate	Display error: "Please enter an integer"	Please enter an integer	Pass
TC20	Copy-paste numeric input	5 (pasted)	1. Copy the value 5 to clipboard 2. Paste into input field 3. Click Calculate	The factorial of 5 is: 120	The factorial of 5 is: 120	Pass
TC21	SQL injection input	1; DROP TABLE	1. Enter 1; DROP TABLE in the input field 2. Click Calculate	Display error: "Please enter an integer"	Please enter an integer	Pass
TC22	Page title verification	N/A	1. Navigate to the application URL 2. Observe the page title in the browser tab and on the page	Page title reads "Factorial!"	Page title reads "Factorial!"	Pass
TC23	Copyright message format	N/A	1. Navigate to the application URL 2. Scroll to the bottom of the page 3. Observe the copyright message	Copyright message shows a valid year (e.g., "2026" or a valid range)	Copyright shows "2026 - 2026" (duplicate year — defect)	Fail

ID	Test Case Description	Input	Test Steps	Expected Result	Actual Result	Status
TC24	Navigation — Terms and Conditions link	N/A	1. Click the "Terms and Conditions" hyperlink	Navigates to the Terms and Conditions page successfully	Page loads without error	Pass
TC25	Navigation — Privacy link	N/A	1. Click the "Privacy" hyperlink	Navigates to the Privacy page successfully	Page loads without error	Pass
TC26	Navigation — About link	N/A	1. Click the "About" hyperlink	Navigates to the About page successfully	Page loads without error	Pass

Notes:

- Boundary testing identifies TC11 (input 170) as the last valid input before JavaScript number overflow occurs at TC12 (input 171).
- TC14 (input ≥ 992) produces no output at all, which is a distinct defect from the Infinity result returned for inputs 171–991.
- Navigation tests (TC24–TC26) verify all three hyperlinks present in the application footer function correctly.